

REVISION 2026

Construction Materials Lab

Diploma in Civil Engineering

Code: 2018

Credits: 1.5

Semester: 2

Course Outcomes (CO)

- **CO1:** Identify the physical properties of bricks and prepare cement mortar. (Apply)
- **CO2:** Identify soil types and flooring tiles. (Apply)
- **CO3:** Identify special construction materials and determine bulking of sand. (Apply)

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

Module 1: Bricks & Mortar

Expt 1: Dimension & Density of Bricks

Measure dimensions of 10 bricks and find average dimension, weight, and density.

Expt 2: Field Test on Bricks

Perform field tests including dropping, striking, and scratching by nail.

Expt 4: Setting out of Building

Setting out a small building in the field using ropes and pegs.

SVG: Brick Dimension Check

Standard Brick (19x9x9 cm)

For the purpose of this experiment, a standard brick is used to check the dimensions of the building. The brick is measured in centimeters (cm) and is used to set out the building. The process of setting out is called *Setting out* and is a common practice in construction.

Module 2: Soil & Flooring

Expt 5: Soil Report

Identify various layers and types of soil in the foundation pit and prepare a report.

Expt 6: Flooring Tiles

Identify different types of flooring tiles such as vitrified, ceramic, glazed, and mosaic tiles.

பொருத்தமான பரப்பை மூடிக்கொடுக்க பொதுவாகப் பயன்படுத்தப்படும் பல்வேறு வகைப் பூக்களை அடையாளம் காணவும். அவைகள் என்னவென்று குறிப்பிடுகவும்.

Module 3: Special Materials & Sand

Expt 8: Bulking of Sand

Determine the increase in volume of sand due to moisture content (Bulking).

பூக்களின் பரப்பை மூடிக்கொடுக்க பொதுவாகப் பயன்படுத்தப்படும் பல்வேறு வகைப் பூக்களை அடையாளம் காணவும். அவைகள் என்னவென்று குறிப்பிடுகவும்.

Practice Model Question Paper

Practice Model Paper — not an official REV2026 paper

1. Find the average weight and density of the given 5 bricks.
2. Prepare a cement mortar of 1:4 proportion for the given quantity of sand.

Source Declaration

Curriculum Scheme: Revision 2026

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Subject Code: 2018

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